

Computing Quasi-Normal Modes
of Schwarzschild Black Holes
with Physics-Informed Neural Networks
Project Onboarding Walkthrough

Jonathan Chung

University of Cambridge

May 10, 2026

Outline

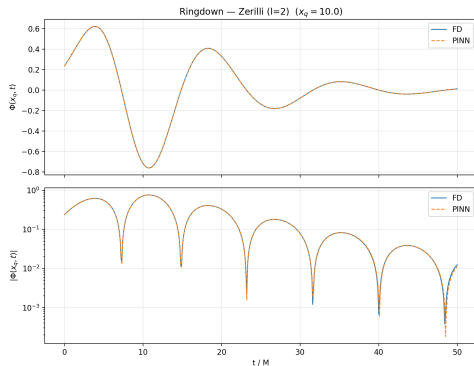
1. Introduction & Background
2. Forward PINN
3. Inverse PINN
4. Curriculum PINN
5. Extraction methods M3, M4, M5
6. Greedy adaptive resampling
7. Summary

The physical picture — black-hole ringdown

A perturbed Schwarzschild black hole “rings” like a struck bell: after the merger, the geometry settles down by emitting **damped sinusoidal** gravitational waves — the **Quasi-Normal Modes (QNMs)**.

$$h(t) \propto e^{-t/\tau} \cos(\omega t + \phi_0)$$

- ω — ringing frequency (real part).
- $\tau = -1/\Im\omega$ — damping time (imaginary part).
- Determined entirely by the black hole’s mass M and spin a .
- Observed by LIGO/Virgo in every binary-black hole detection since GW150914.



Ringdown waveform $\Phi(x_q=10M, t)$

From Einstein's equations to a 1+1D wave problem

Step 1 — background. Schwarzschild geometry (M is the black hole mass),

$$ds^2 = -f(r) dt^2 + f(r)^{-1} dr^2 + r^2 d\Omega^2, \quad f(r) = 1 - \frac{2M}{r}.$$

Step 2 — linear perturbation. Write $g_{\mu\nu} = g_{\mu\nu}^{(0)} + h_{\mu\nu}$ with $|h| \ll |g^{(0)}|$ and expand the vacuum Einstein equations to first order:

$$\delta R_{\mu\nu}[h] = 0.$$

Step 3 — exploit spherical symmetry. Decompose $h_{\mu\nu}$ in *tensor spherical harmonics*; under $(\theta, \varphi) \rightarrow (\pi - \theta, \varphi + \pi)$ the modes split into two **decoupled parity sectors**:

Sector	Parity	Master equation
Axial (odd)	$(-1)^{\ell+1}$	Regge–Wheeler
Polar (even)	$(-1)^{\ell}$	Zerilli

Each sector reduces to a single scalar field $\Phi(t, r_*)$ on the **tortoise coordinate** $dr_* = dr/f(r)$.

The two master equations — Regge–Wheeler and Zerilli

Both sectors obey the **same 1+1D wave form**

$$-\frac{\partial^2 \Phi}{\partial t^2} + \frac{\partial^2 \Phi}{\partial r_*^2} - V_\ell(r) \Phi = 0,$$

but with two *different* effective potentials:

Regge–Wheeler (axial)

$$V_\ell^{\text{RW}}(r) = f(r) \left[\frac{\ell(\ell+1)}{r^2} - \frac{6M}{r^3} \right]$$

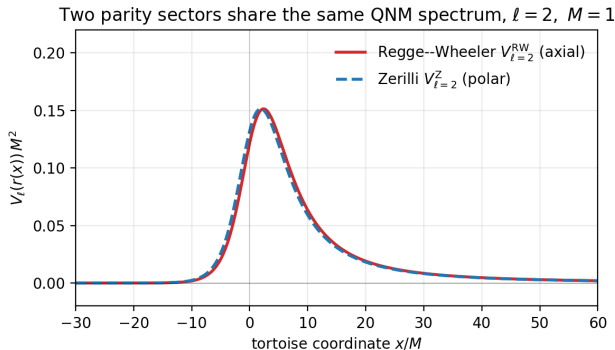
Zerilli (polar)

$$V_\ell^{\text{Z}}(r) = f(r) \frac{2[n^2(n+1)r^3 + 3n^2Mr^2 + 9nM^2r + 9M^3]}{r^3(nr + 3M)^2}$$

$$n = \frac{1}{2}(\ell - 1)(\ell + 2).$$

Two *different* potentials, one for each parity sector — but as we will see on the next slide, they encode *the same physics*.

Isospectrality — one spectrum, two potentials



Chandrasekhar–Detweiler isospectrality: the two potentials are *distinct* but share *exactly the same* QNM spectrum $\{\omega_n\}$.

$\Rightarrow$ Either equation is a valid target for our solver.

This work uses the **Zerilli (polar, $\ell=2$)** form, matching the reference PINN literature (Patel *et al.* 2024) and giving a single, well-defined benchmark to push accuracy against.

The numerical problem we actually solve

Target: the polar $\ell=2$ Zerilli equation on $r_*/M \in [-50, 150]$, $t/M \in [0, 50]$.

- **Initial condition** ($t = 0$): localised Gaussian pulse, time-symmetric ($\partial_t \Phi = 0$).
- **Boundary conditions:** *purely outgoing* waves at $r_* \rightarrow +\infty$ (radiation to infinity), *purely ingoing* at $r_* \rightarrow -\infty$ (absorption by the horizon).
- These BCs are precisely what selects the discrete **QNM spectrum** $\omega_n \in \mathbb{C}$.

Reference fundamental mode ($\ell=2$, $n=0$, Leaver 1985):

$$M\omega = 0.3737 - 0.0890 i, \quad \tau/M = -1/\Im\omega = 11.241.$$

Every result in the rest of the talk is benchmarked against this number.

Three ways to compute QNMs

1. Leaver continued fraction (1985) — the *exact reference*

Frobenius series ansatz $\Rightarrow$ 3-term recurrence $\Rightarrow$ Pincherle continued fraction.

+ 12-digit accuracy in milliseconds. – only works on known analytic potentials.

2. Finite-difference (FD) time evolution — the *ground truth waveform*

2nd-order centred FD on the (r_*, t) grid; extract ω, τ from the late-time signal.

+ produces the full (x, t) field used to train our PINN. – slow.

3. Physics-Informed Neural Network (PINN) — our tool

NN $\Phi_\theta(x, t)$ trained to satisfy PDE + IC + BC.

+ mesh-free, differentiable, jointly learns physical parameters. – slower and less accurate than Leaver/FD on Schwarzschild.

Why bother with PINNs at all on Schwarzschild?

- Schwarzschild is the **controlled benchmark**: every answer is known to 12 digits, so any PINN error is *measurable*.
- PINNs are **differentiable end-to-end**: the same training loop can solve the *inverse* problem (recover M, ω, τ from data).
- Patel et al. 2024 introduced PINNs for QNMs but report only ~ 1 – 2 digit accuracy.
- **Our work pushes this methodology into a regime where it actually competes** with classical extractors.

What this project actually contributes

Schwarzschild $\ell=2$ is the **stress test** for QNM-PINNs — every error is measurable against an exact reference. Our contributions:

- ➊ **Forward PINN, accuracy push:** hard boundary conditions, gPINN residuals, **greedy adaptive sampling**, extended L-BFGS. Pushes spatial RMSD down by $\sim 55\%$ vs. uniform sampling and ω error to 0.03%.
- ➋ **Inverse PINN:** learn M, ω, τ from ringdown data in a single training run. Best variant (**D_combo**): M to 0.34%, ω to 0.31%, τ to 1.6%.
- ➌ **Curriculum PINN:** split $t \in [0, 50]M$ into K windows trained sequentially, IC of window k taken from network $k-1$. Diagnoses where the late-time error of the single-window forward PINN comes from.
- ➍ **New extraction methodology — M3, M4, M5:** ESPRIT subspace decomposition; two-mode NLS with 1-D and 2-D *plateau scans* to give principled, falsifiable (ω, τ) estimates with systematic-error bars.
- ➎ **Greedy resampling:** simple, deterministic, hyperparameter-light replacement for RAD with measurably better accuracy.

Forward PINN — problem statement

Goal. Train a neural network $\Phi_{\theta}(x, t)$ to satisfy the Zerilli PDE on the full (x, t) domain, plus its initial and boundary conditions.

We minimise a sum of **seven** weighted mean-squared-error terms,

$$\mathcal{L}_{\text{fwd}}(\theta) = \underbrace{\lambda_r \mathcal{L}_r + \lambda_{r_x} \mathcal{L}_{r_x} + \lambda_{r_t} \mathcal{L}_{r_t}}_{\text{PDE residual} + \text{gPINN}} + \underbrace{\lambda_{ic} \mathcal{L}_{ic} + \lambda_{iv} \mathcal{L}_{iv}}_{\text{initial condition at } t=0} + \underbrace{\lambda_{bl} \mathcal{L}_{bl} + \lambda_{br} \mathcal{L}_{br}}_{\text{boundary conditions}}$$

Each $\mathcal{L}_i$ is a mean of squared residuals over a sample of collocation points; the next two slides write out all seven sums explicitly.

Forward PINN loss — PDE residual + gPINN

Define the **PDE residual** of the Zerilli equation,

$$r_\theta(x, t) \equiv -\partial_t^2 \Phi_\theta(x, t) + \partial_x^2 \Phi_\theta(x, t) - V_\ell(r) \Phi_\theta(x, t).$$

At $N_r = 32,000$ randomly drawn collocation points $\{(x_i, t_i)\}_{i=1}^{N_r}$ inside the domain we evaluate three mean-squared losses:

$$\mathcal{L}_r = \frac{1}{N_r} \sum_{i=1}^{N_r} [r_\theta(x_i, t_i)]^2 \quad (\text{standard PINN residual})$$

$$\mathcal{L}_{r_x} = \frac{1}{N_r} \sum_{i=1}^{N_r} [\partial_x r_\theta(x_i, t_i)]^2, \quad \mathcal{L}_{r_t} = \frac{1}{N_r} \sum_{i=1}^{N_r} [\partial_t r_\theta(x_i, t_i)]^2$$

The two extra terms are the **gPINN** regularisers of Yu *et al.* 2022: penalising the gradient of the residual forces the residual to be *smoothly* small, not just point-wise small, and gives noticeably cleaner derivatives.

Forward PINN loss — initial and boundary conditions

Initial condition at $t = 0$, on N_{ic} points $\{x_j\}_{j=1}^{N_{ic}}$ along the spatial axis:

$$\mathcal{L}_{ic} = \frac{1}{N_{ic}} \sum_{j=1}^{N_{ic}} [\Phi_{\theta}(x_j, 0) - \Phi_0(x_j)]^2 \quad \Phi_0(x) \text{ is a Gaussian pulse,}$$

$$\mathcal{L}_{iv} = \frac{1}{N_{ic}} \sum_{j=1}^{N_{ic}} [\partial_t \Phi_{\theta}(x_j, 0)]^2 \quad (\text{time-symmetric initial data}).$$

Boundary conditions, on N_{bc} time points $\{t_k\}_{k=1}^{N_{bc}}$ at each domain edge — outgoing waves at $x_{\max}$, ingoing at $x_{\min}$:

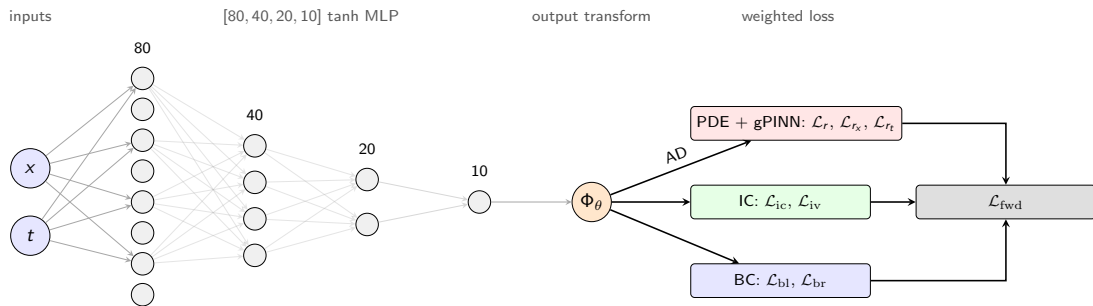
$$\mathcal{L}_{bl} = \frac{1}{N_{bc}} \sum_{k=1}^{N_{bc}} [(\partial_t - \partial_x) \Phi_{\theta}(x_{\min}, t_k)]^2,$$

$$\mathcal{L}_{br} = \frac{1}{N_{bc}} \sum_{k=1}^{N_{bc}} [(\partial_t + \partial_x) \Phi_{\theta}(x_{\max}, t_k)]^2.$$

Forward PINN — training setup

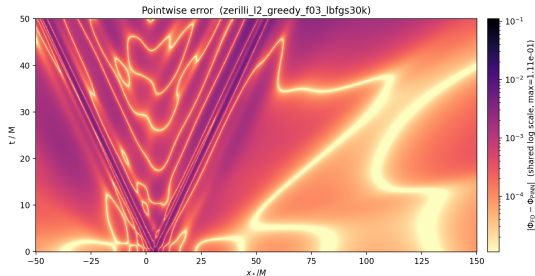
- **Network.** Fully-connected MLP with layer widths $[2, 80, 40, 20, 10, 1]$, tanh activations ($\sim 4,500$ trainable parameters).
- **Output transform.** $\Phi_\theta(x, t) = A \tanh(\tilde{\Phi}_\theta(x, t))$ with $A = 1$, where $\tilde{\Phi}_\theta$ is the raw network output. This bounds the field and stabilises early training.
- **Loss weights.** $\lambda = (\lambda_r, \lambda_{r_x}, \lambda_{r_t}, \lambda_{ic}, \lambda_{iv}, \lambda_{bl}, \lambda_{br}) = (100, 100, 100, 1, 100, 1, 1)$.
- **Optimiser.** Adam for 10,000 steps to find a good basin, then **L-BFGS** for 30,000 steps to converge inside it.
- **Sampling.** Collocation points are re-drawn every epoch via the *greedy resampling* scheme described in §6.

Forward PINN — our architecture

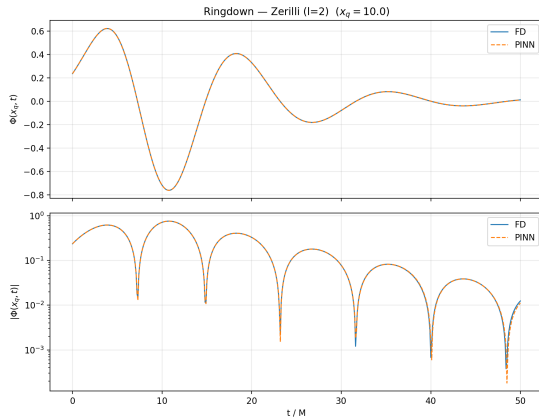


Inputs (x, t) feed a four-layer tanh MLP of widths $80 \rightarrow 40 \rightarrow 20 \rightarrow 10$ ($\sim 4,500$ params); the scalar output passes through $\Phi_\theta = A \tanh(\tilde{\Phi}_\theta)$ before being differentiated by automatic differentiation (**AD**) to form the seven residuals of the previous two slides.

Forward PINN — best result: waveform agreement



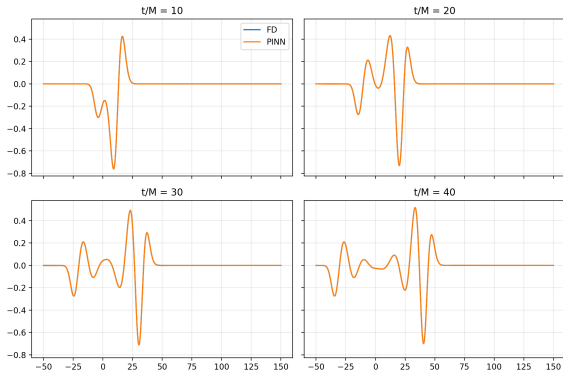
Pointwise error $|\Phi_{PINN} - \Phi_{FD}|$ over the full (x, t) domain.



Ringdown overlay at $x_q = 10 M$: PINN (solid) vs. FD (dashed). Greedy resampling, $f=0.3$, + L-BFGS 30k.

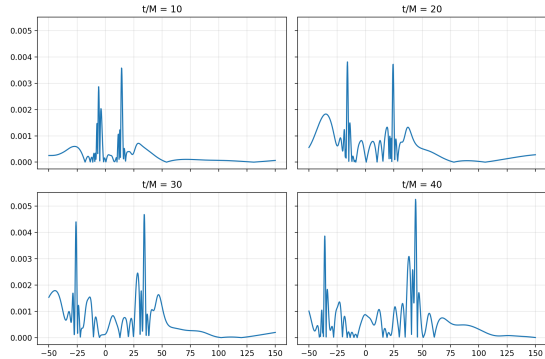
Forward PINN — waveform snapshots and pointwise error

Field $\Phi(x, t)$ snapshots — Zerilli $l = 2$ (FD vs PINN)



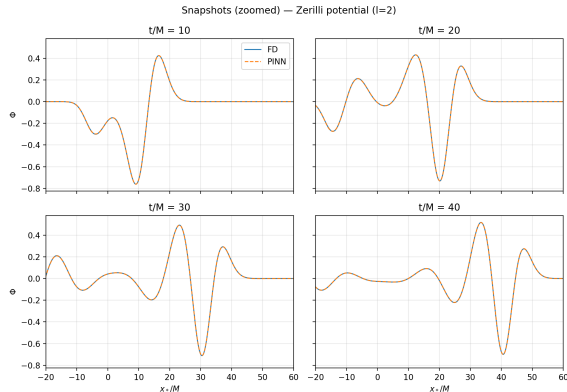
Φ_{PINN} (solid) vs. Φ_{FD} (dashed) at $t/M \in \{10, 20, 30, 40\}$.

Absolute difference — Zerilli potential ($l=2$)

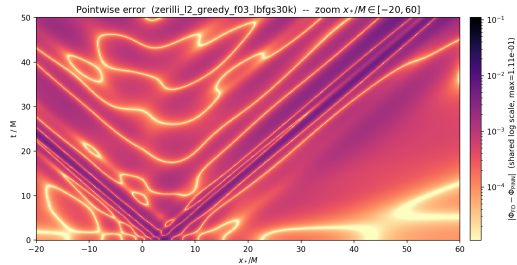


Pointwise $|\Phi_{PINN} - \Phi_{FD}|$ at the same times: residual
 $\lesssim 5 \times 10^{-3}$ near the barrier, $\lesssim 10^{-3}$ far away.

Forward PINN — zoomed late-time waveform and error



Zoomed snapshots near the potential barrier; the PINN tracks both phase and decay envelope of the FD ground truth.



Zoomed pointwise error: residual concentrated on the prompt burst $t \lesssim 15M$, near-uniform during the ringdown.

Forward PINN — best result: QNM accuracy

Quantity	Our value	Leaver reference	Error
RMSD vs. FD	8.17×10^{-4}	—	—
ω (M4 plateau)	0.37359	0.37367	0.029%
τ (M2, window $[10, 50]M$)	11.46	11.241	1.99%

Patel et al. 2024 report $\sim 1\%$ accuracy on ω .
We reach 0.03% — a $\sim 30\times$ improvement.

Inverse PINN — problem statement

Forward problem: given M (hence $V(r; M)$), solve for the field $\Phi(x, t)$.

Inverse problem: given an observed waveform $\Phi_{FD}(x, t)$, recover the underlying physical parameters (M, ω, τ) .

We minimise a sum of **nine** weighted mean-squared-error terms: the seven forward losses (with V now evaluated at the *learned* mass M) plus two new data-driven terms,

$$\mathcal{L}_{\text{inv}} = \underbrace{\mathcal{L}_{\text{PDE+gPINN}}(M)}_{\text{Zerilli residual at learned } M} + \underbrace{\mathcal{L}_{\text{IC+BC}}}_{\text{geometry}} + \underbrace{\lambda_{\text{data}} \mathcal{L}_{\text{data}}}_{\text{noisy waveform fit}} + \underbrace{\lambda_{\text{ring}} \mathcal{L}_{\text{ring}}}_{\text{ringdown-template fit}}$$

Five physical scalars $(M, \omega, \tau, A_{\text{ring}}, \phi)$ are appended to the optimiser's parameter list and trained *jointly* with the network weights θ . The next slide writes the two new sums explicitly.

Inverse PINN loss — data and ringdown terms

Data loss on $N_{\text{obs}}=500$ randomly drawn observation points $\{(x_j, t_j)\}_{j=1}^{N_{\text{obs}}}$ with FD reference values $\Phi_{FD}(x_j, t_j)$:

$$\mathcal{L}_{\text{data}} = \frac{1}{N_{\text{obs}}} \sum_{j=1}^{N_{\text{obs}}} [\Phi_{\theta}(x_j, t_j) - \Phi_{FD}(x_j, t_j)]^2.$$

Ringdown-template loss at the observer column $x_q=10 M$, on N_{ring} time samples $\{t_k\}_{k=1}^{N_{\text{ring}}}$ in the late-time window $t_k \in [t_{\text{ring}}^{\min}, t_{\text{max}}]$:

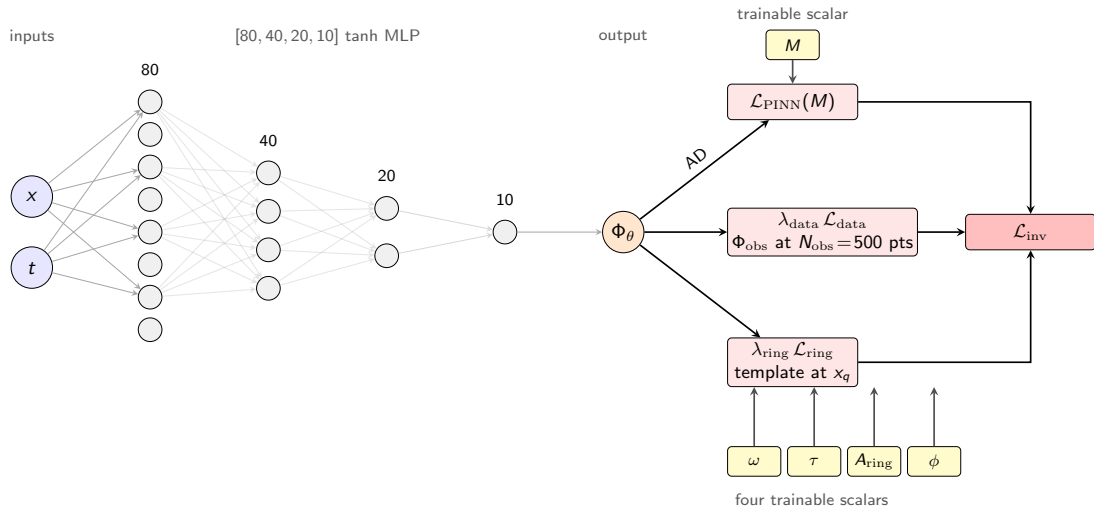
$$\mathcal{L}_{\text{ring}} = \frac{1}{N_{\text{ring}}} \sum_{k=1}^{N_{\text{ring}}} [\Phi_{\theta}(x_q, t_k) - A_{\text{ring}} e^{-t_k/\tau} \cos(\omega t_k + \phi)]^2.$$

Through this sum, the four template scalars $(\omega, \tau, A_{\text{ring}}, \phi)$ receive gradients directly from the late-time portion of Φ_{θ} , while M receives gradients through $\mathcal{L}_{\text{PDE+gPINN}}(M)$ via $V(r; M)$.

Inverse PINN — training setup

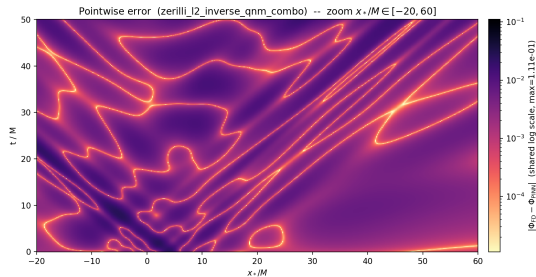
- **Network.** Identical $[2, 80, 40, 20, 10, 1]$ tanh MLP and output transform as the forward PINN.
- **Learnable physical parameters** (initial values): $M_{\text{init}} = 1.20$ (twenty per cent above the truth), $\omega_{\text{init}} = 0.30$, $\tau_{\text{init}} = 8.00$, $A_{\text{ring, init}}$ and ϕ_{init} from a short pre-fit on the FD waveform.
- **Loss weights** for the two new terms: $\lambda_{\text{data}} = 1$, $\lambda_{\text{ring}} = 100$. Forward weights unchanged.
- **Optimiser.** Adam (10,000 steps) followed by L-BFGS (30,000 steps); the five physical scalars and the network weights are updated together in every step.
- **Variant D combo:** $t_{\text{ring}}^{\text{min}} = 18 M$, $N_{\text{ring}} = 1000$, late-time window only — our best configuration.

Inverse PINN — our architecture

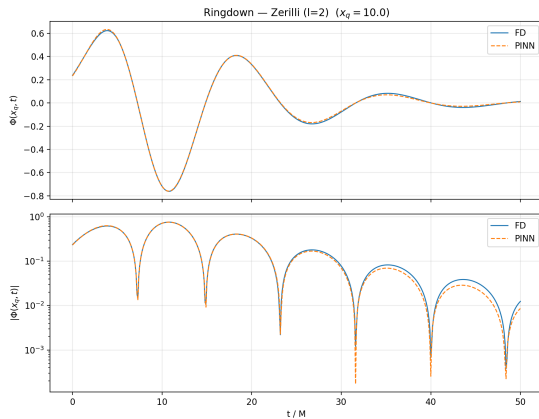


Same [80, 40, 20, 10] tanh MLP as the forward PINN, with five physical scalars ($M, \omega, \tau, A_{\text{ring}}, \phi$) (yellow) appended to the optimiser's parameter list. M enters the PDE residual through $V(r; M)$; $(\omega, \tau, A_{\text{ring}}, \phi)$ enter $\mathcal{L}_{\text{ring}}$ at the observer column $x_q = 10 M$.

Inverse PINN — best result (variant D “combo”): waveform



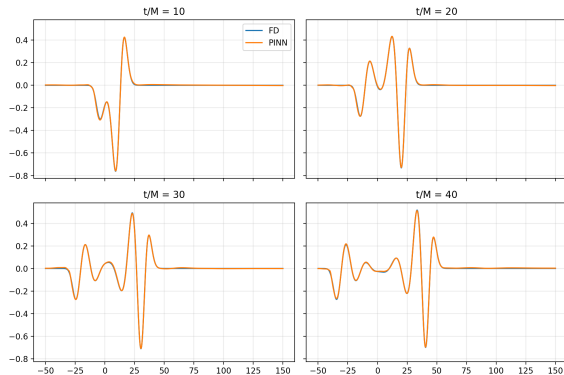
$|\Phi_\theta - \Phi_{FD}|$ on the zoomed late-time region.



PINN ringdown at $x_q = 10 M$ vs. FD ground truth.

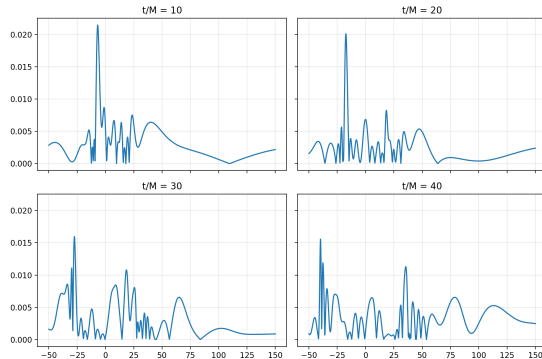
Inverse PINN — waveform snapshots and pointwise error

Inverse+QNM PINN — Zerilli ($l=2$), $M=1.0034$



Inverse PINN Φ_θ (solid) vs. Φ_{FD} (dashed) at $t/M \in \{10, 20, 30, 40\}$, with M treated as unknown.

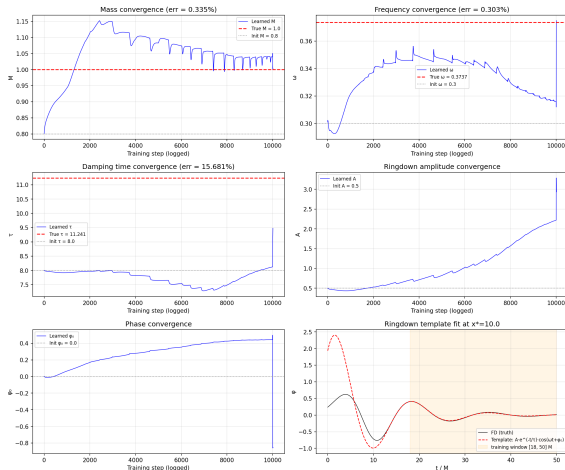
Abs difference — Zerilli ($l=2$)



Pointwise $|\Phi_\theta - \Phi_{FD}|$ at the same times; waveform recovery comparable to the forward run while M is simultaneously inferred to 0.34%.

Inverse PINN — parameter convergence

Inverse+QNM Parameter Convergence (noise=1.0%)



Trajectories of the trainable scalars during Adam $\rightarrow$ L-BFGS:

- M : $1.20 \rightarrow 1.0034$
- ω : $0.30 \rightarrow 0.3748$
- τ : $8.00 \rightarrow 11.06$
- A_{ring}, ϕ co-fit on the $[t_{\text{ring}}^{\min}, t_{\text{max}}]$ window.

All five physical scalars settle inside the L-BFGS phase, with no observed parameter drift after convergence.

Inverse PINN — best result: recovered parameters

Parameter	Init	Learned	True	Error
M	1.20	1.0034	1.0	0.34%
ω (M4 plateau)	0.30	0.3748	0.3737	0.31%
τ (M2, window $[10, 50]M$)	8.00	11.06	11.241	1.59%

Variant D combo hyperparameters: $t_{\text{ring}}^{\min} = 18M$, $\lambda_{\text{ring}} = 100$, $N_{\text{ring}} = 1000$.
All three physical parameters recovered to $\leq 1.6\%$ in a single training run.

Curriculum PINN — problem statement

Diagnosis. The forward PINN's pointwise error grows monotonically with t : the late-time band $t \in [37.5, 50]M$ contributes $\sim 41\%$ of the full-domain RMSD, despite spanning only a quarter of the temporal domain.

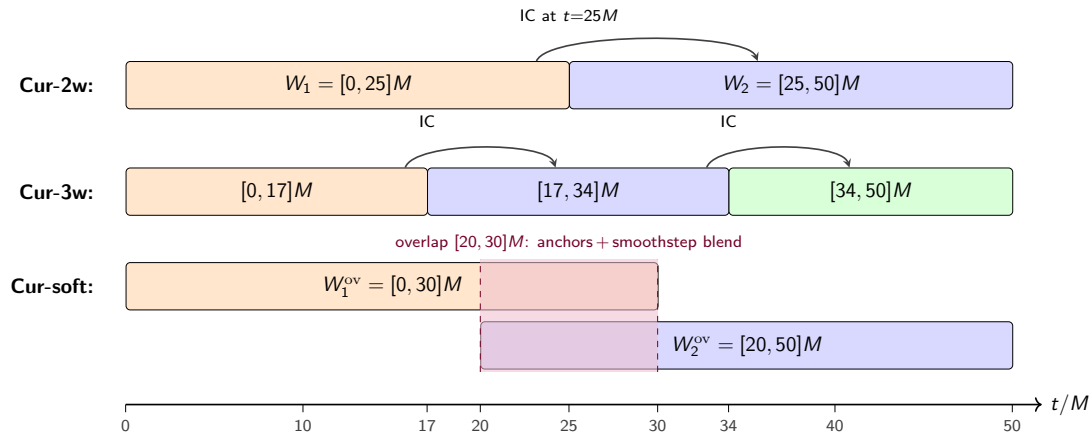
Idea. Replace the single-window forward PINN with a *curriculum* [Bengio et al. 2009; Krishnapriyan et al. 2021]: split $t \in [0, 50]M$ into K contiguous windows $W_k = [t_{k-1}, t_k]$ and train an *independent* PINN Φ_{θ_k} on each, with the network of window k supplying the initial conditions of window $k+1$:

$$\Phi_k^{\text{IC}}(x) = \Phi_{\theta_{k-1}}(x, t_{k-1}), \quad \partial_t \Phi_k^{\text{IC}}(x) = \partial_t \Phi_{\theta_{k-1}}(x, t_{k-1}).$$

Trade-off.

- **Pro:** smaller temporal sub-domain $\Rightarrow$ easier optimisation landscape per window.
- **Con:** every window after the first inherits the residual error of its predecessor through its IC.

Curriculum PINN — three partitions



Cur-2w and **Cur-3w**: hard partition — IC of each window extracted from the previous network at the splice time.
Cur-soft: overlapping windows with continuity anchors and a smoothstep blend (next slide).

Curriculum PINN — loss per window

Each window W_k minimises the **same seven-term forward loss** as the forward PINN, evaluated on W_k only:

$$\mathcal{L}_{\text{cur},k} = \underbrace{\lambda_r \mathcal{L}_r^{(k)} + \lambda_{r_x} \mathcal{L}_{r_x}^{(k)} + \lambda_{r_t} \mathcal{L}_{r_t}^{(k)}}_{\text{PDE residual + gPINN on } W_k} + \underbrace{\lambda_{\text{ic}} \mathcal{L}_{\text{ic}}^{(k)} + \lambda_{\text{iv}} \mathcal{L}_{\text{iv}}^{(k)}}_{\text{IC at } t=t_{k-1}} + \underbrace{\lambda_{\text{bl}} \mathcal{L}_{\text{bl}}^{(k)} + \lambda_{\text{br}} \mathcal{L}_{\text{br}}^{(k)}}_{\text{boundary conditions}}$$

Initial condition losses differ by window:

- **Window 1** ($k=1$): same analytic Gaussian-pulse IC as the forward PINN,
 $\mathcal{L}_{\text{ic}}^{(1)} = \frac{1}{N_{\text{ic}}} \sum_j [\Phi_{\theta_1}(x_j, 0) - \Phi_0(x_j)]^2$, $\mathcal{L}_{\text{iv}}^{(1)} = \frac{1}{N_{\text{ic}}} \sum_j [\partial_t \Phi_{\theta_1}(x_j, 0)]^2$.
- **Window** $k \geq 2$: IC extracted from the *trained* previous network $\Phi_{\theta_{k-1}}$ at the splice time t_{k-1} :

$$\mathcal{L}_{\text{ic}}^{(k)} = \frac{1}{N_{\text{ic}}} \sum_j [\Phi_{\theta_k}(x_j, t_{k-1}) - \Phi_{\theta_{k-1}}(x_j, t_{k-1})]^2$$

$$\mathcal{L}_{\text{iv}}^{(k)} = \frac{1}{N_{\text{ic}}} \sum_j [\partial_t \Phi_{\theta_k}(x_j, t_{k-1}) - \partial_t \Phi_{\theta_{k-1}}(x_j, t_{k-1})]^2$$

All weights λ and the (Adam 10,000 + L-BFGS 30,000) schedule are identical to the forward PINN, applied independently per window.

Curriculum PINN — soft-overlap variant

Two windows share an overlap band of width Δ_{ov} centred on t_{split} :

$$W_1^{\text{ov}} = [t_{\min}, t_{\text{split}} + \Delta_{\text{ov}}/2], \quad W_2^{\text{ov}} = [t_{\text{split}} - \Delta_{\text{ov}}/2, t_{\max}].$$

Continuity-anchor loss on a fixed $n_x \times n_t$ grid spanning the overlap rectangle:

$$\mathcal{L}_{\text{cont}} = \frac{\lambda_{\text{cont}}}{n_x n_t} \sum_{i,j} (\Phi_{\theta_2}(x_i, t_j) - \Phi_{\theta_1}(x_i, t_j))^2,$$

added to Φ_{θ_2} 's loss; pulls window 2 towards the already-trained window 1 across the entire overlap rather than at the splice time alone.

Smoothstep blend on the overlap band:

$$\Phi_{\text{cur}}^{\text{ov}}(x, t) = (1 - s(\xi)) \Phi_{\theta_1} + s(\xi) \Phi_{\theta_2}, \quad s(\xi) = \xi^2(3 - 2\xi), \quad \xi = \frac{t - t_{\text{ov}}^-}{\Delta_{\text{ov}}}.$$

s is C^1 with $s(0)=0$, $s(1)=1$, $s'(0)=s'(1)=0 \Rightarrow$ stitched field is continuously differentiable across the overlap edges.

Settings. $\Delta_{\text{ov}} = 10M$, $n_x \times n_t = 201 \times 11$, $\lambda_{\text{cont}} = 10$.

Curriculum PINN — variants and results

Variant	Full-domain RMSD ($\times 10^{-3}$)	W_1 RMSD ($\times 10^{-3}$)	W_2 RMSD ($\times 10^{-3}$)	W_3 /overlap RMSD ($\times 10^{-3}$)
Baseline (single window)	0.817	—	—	—
Cur-2w	0.845	0.516	1.077	—
Cur-3w	0.976	0.554	0.920	1.322
Cur-soft	3.155	1.739	3.932	2.394 (overlap)

- **Cur-2w ties baseline** on full domain (+3%); W_1 alone reaches 5.16×10^{-4} (−37% vs. baseline on the same sub-domain) — the predicted local benefit *is* delivered.
- **Cur-3w**: per-window RMSD grows monotonically ($5.5 \rightarrow 9.2 \rightarrow 13.2$); each splice adds IC error. Full-domain +19%.
- **Cur-soft fails**: $\sim 4\times$ baseline RMSD; M4 plateau scan finds no contiguous block satisfying the relative-scatter criterion.

Take-away. Curriculum buys local accuracy on W_1 but pays it back through IC-transfer error on later windows; no net gain over the single-window forward PINN.

Why we need new extraction methods

Patel et al. 2024 (and our earlier work) used:

- **M1 — FFT + log-linear envelope:** ω from FFT peak, τ from $\log |\text{peaks}|$ slope.
- **M2 — single-mode NLS curve fit:** fit $Ae^{-t/\tau} \cos(\omega t + \phi)$ over a chosen window.

Two unsolved problems with M1/M2:

- ① **Overtone contamination.** The first overtone $\tau_1 = 3.67 M$ is comparable to the burst at early times. A single-mode fit absorbs it as bias.
- ② **Window dependence.** M1 and M2 quote a single number from a single window, with no estimate of how that number changes if the window edges move. **This is not a falsifiable measurement.**

M3, M4, M5 are our advances that address both.

M3 — ESPRIT subspace decomposition

Idea: treat the ringdown as a sum of K complex exponentials and solve for them *simultaneously* using subspace methods (Estimation of Signal Parameters via Rotational Invariance Techniques).

$$\Phi(x_q, t) \approx \sum_{k=1}^K A_k e^{(i\omega_k - 1/\tau_k) t}$$

Procedure:

- 1 Optionally Hilbert-transform the real signal to its analytic (complex) form so each physical mode appears once.
- 2 Build the Hankel data matrix; SVD to get the signal subspace.
- 3 Solve the rotational-invariance eigenvalue problem $\mathbf{U}_1^\dagger \mathbf{U}_2 = \boldsymbol{\Psi}$.
- 4 Eigenvalues of $\boldsymbol{\Psi} \rightarrow (\omega_k, \tau_k)$; report the largest-amplitude mode.

Strength: no initial guess; resolves multiple modes naturally.

Weakness: sensitive to Hankel-matrix conditioning at low SNR; on inverse-PINN waveforms can become ill-conditioned.

M4 — two-mode NLS with start-time plateau scan

Step 1. Fit a *two-mode* template (fundamental + first overtone):

$$\Phi_{\text{model}}(t) = A_0 e^{-t/\tau_0} \cos(\omega_0 t + \phi_0) + A_1 e^{-t/\tau_1} \cos(\omega_1 t + \phi_1)$$

using `scipy.optimize.curve_fit` with bound constraints, on window $[t_0, t_{\text{max}}]$. Identify the longer- τ mode as the fundamental.

Step 2. Repeat over a grid of start times $t_0 \in [10, 25]M$. Identify the contiguous **plateau** of low-scatter consecutive fits.

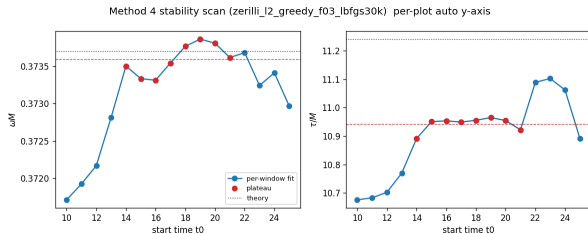
Step 3. Report:

$$\omega_{M4} = \langle \omega \rangle_{\text{plateau}}, \quad \sigma_{\omega} = \text{std}(\omega)_{\text{plateau}}$$

— the plateau spread is a **principled systematic error bar**.

Why it works: an honest QNM extraction must be invariant to small shifts of the fit window; the plateau is the empirical signature of that invariance.

M4 — example stability scan



Forward PINN: M4 stability scan over $t_0 \in [10, 25]M$

- ω (top): essentially flat across most of $t_0 \in [14, 21]M \Rightarrow$ plateau identified.
- τ (bottom): converges to a stable value once t_0 is past the prompt burst.
- Plateau quote:
 $\omega_{M4} = 0.37359 \pm 1.96 \times 10^{-4} \Rightarrow 0.029\%$ error vs. Leaver.
- Vertical span of the green band $= 2\sigma_\omega$ — an honest, falsifiable systematic.

M5 — two-mode NLS with 2-D plateau scan

M4 sweeps only one window edge (t_0). **M5** sweeps both:

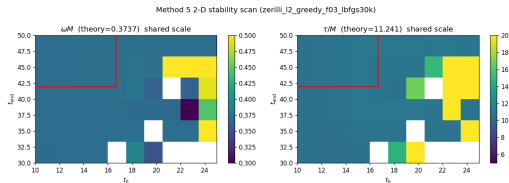
$$[t_0, t_{\text{end}}] \in [10, 25]M \times [t_{\text{min}}, 50]M \quad \text{on a } 10 \times 6 \text{ grid.}$$

Identify the contiguous **rectangular plateau** of 5×3 with smallest combined relative scatter; report plateau-mean and plateau-stddev.

Why this matters:

- t_0 scans control *overtone leakage* (early edge).
- t_{end} scans control *Price tail leakage* (late edge).
- Truncating either edge prematurely is a model-misspecification bias, not a noise issue — it does *not* go away with more data.
- M5 is, to our knowledge, the **first systematic 2-D plateau analysis** applied to PINN-derived QNM extraction.

M5 — example 2-D heatmap



Top: ω scatter map. **Bottom:** τ scatter map.

- x-axis: t_0 (start of fit window).
- y-axis: t_{end} (end of fit window).
- Colour: relative scatter inside the local rectangle (capped at 5%).
- **Red box:** plateau (minimum joint scatter).
- Grey: NaN cells where the two-mode fit failed.

Plateau quote:

$$\omega_{M5} = 0.3732 \text{ (0.13\% bias)}, \quad \tau_{M5} = 10.82 \text{ (3.91\% bias)}.$$

M5 reproduces M4 *and* certifies the right-edge choice is harmless — a property M4 alone cannot test.

Reporting convention — best estimator per quantity

Different physical quantities are best estimated from different features of the waveform. We therefore quote a **hybrid headline**:

Quantity	Estimator	Reason
ω	M4 plateau mean	Phase information aggregated across cycles; lowest-variance estimator ($\sigma_\omega \sim 10^{-5}$).
τ	M2 over wide window $[10, 50]M$	Amplitude information over longest decay range; avoids early-edge overtone bias.

- This is consistent with established practice in numerical-relativity ringdown analysis (Berti et al. 2007).
- Patel et al. 2024 use a similar split (FFT for ω , envelope for τ) but report *no fit-window sensitivity*.
- M3, M4, M5 are also reported as cross-checks; on healthy forward runs all three agree to within their plateau widths.

All extraction methods on the best forward PINN

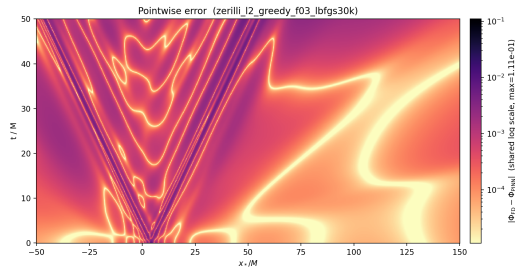
Method	ω	ω err	τ	τ err
Truth (Leaver)	0.3737	—	11.241	—
M1 (FFT + envelope)	0.3712	0.68%	10.92	2.83%
M2 (single-mode NLS)	0.3794	1.54%	11.46	1.99%
M3 (ESPRIT, $K=6$)	0.4829	29.2%	7.84	30.3%
M4 (two-mode plateau)	0.37359	0.029%	10.94	2.65%
M5 (2-D plateau)	0.37284	0.23%	10.80	3.91%

- **M4 wins on ω by an order of magnitude** — plateau scan + two-mode template removes both overtone bias and window-choice arbitrariness.
- **M3 fails** on this run: ESPRIT's Hankel matrix is ill-conditioned at low signal levels (max amp $\sim 10^{-24}$ flagged automatically).
- M2 wide-window remains the best τ estimator.
- All methods are reported *together* as the cross-check — divergences flag waveform pathologies.

Why adaptive sampling matters

Uniform collocation wastes points where the residual is already small and starves the network where it is large.

- **Residual-Adaptive Distribution (RAD)** [Wu et al. 2023]: probabilistic resampling with $p_i \propto |r_i|^k + c$. Two hyperparameters (k, c).
- Stochastic $\Rightarrow$ extreme residuals are not always selected.
- Sampling noise can interfere with Adam dynamics.



Final error map: PINN residual concentrates along the wavefronts that travel out from the burst.

Greedy resampling — algorithm

Replace probabilistic RAD with **deterministic top- N** selection:

- 1 Every P Adam steps: generate $N_{\text{cand}}=100,000$ random candidates in the domain.
- 2 Evaluate the PDE residual $|r_i|$ at each candidate (one extra forward pass — batched, $\ll$ training cost).
- 3 **Sort by $|r_i|$ descending**; keep the top $f \cdot N_r$ points.
- 4 Fill the remaining $(1-f) \cdot N_r$ slots with **uniform random** points (preserves exploration).
- 5 Replace all $N_r=32,000$ collocation points; continue Adam.

Key advantages over RAD:

- **Deterministic** — the worst residuals are always kept.
- **Single hyperparameter f** . $f=0$: uniform; $f=1$: fully greedy.
- No need to tune the $|r|^k + c$ shape.

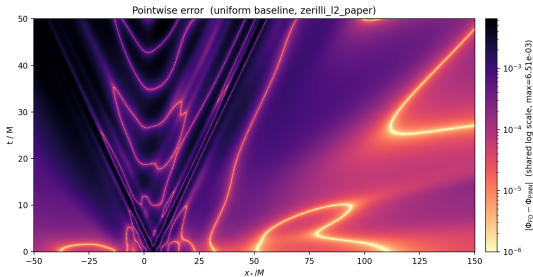
Greedy resampling — f sweep results

Configuration	RMSD	vs. baseline
Uniform (baseline)	0.00183	—
RAD $k=2$	0.00095	−48%
Greedy $f=0.3$	0.00085	−54%
Greedy $f=0.5$	0.00107	−42%
Greedy $f=0.8$	0.00165	−10%
Greedy $f=0.3$ + L-BFGS 30k	0.00082	−55%

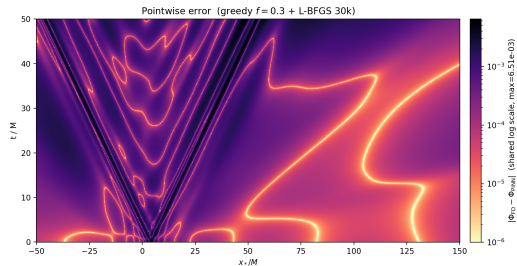
- $f=0.3$ **optimal**: 30% greedy + 70% uniform; balances exploitation and exploration.
- Too aggressive ($f \geq 0.5$): network over-fits the high-residual region and regresses elsewhere.
- **Beats RAD with simpler algorithm.**
- Combined with extended L-BFGS: best forward PINN result.

One knob, deterministic, beats RAD.
Now the default for all forward and inverse PINN runs.

Greedy resampling — error map: uniform vs. greedy $f=0.3$



Uniform baseline (Patel et al. 2024 reproduction): residual concentrates along the outgoing wavefronts.



Greedy $f=0.3$ + L-BFGS 30k: same colour scale, pointwise error reduced by an order of magnitude in the wavefront band.

Summary of the project

- 1 **Problem:** compute Schwarzschild $\ell=2$ QNMs from the Zerilli PDE using PINNs — forward and inverse.
- 2 **Forward PINN best:** greedy $f=0.3$ + L-BFGS 30k.
 ω to **0.029%**, τ to **1.99%**, RMSD -55% vs. uniform baseline.
- 3 **Inverse PINN best:** variant D (combo).
 M to **0.34%**, ω to **0.31%**, τ to **1.59%** from a single training run.
- 4 **Extraction methodology M3, M4, M5:**
Two-mode templates + plateau scans give principled, falsifiable (ω, τ) estimates with systematic-error bars; M5 is the first systematic 2-D plateau analysis for PINN-derived ringdowns.
- 5 **Greedy resampling:** simple, deterministic, one-hyperparameter improvement on RAD; now the default.

Where we are vs. where we go next

Honest scope:

- PINNs cannot beat Leaver on forward Schwarzschild — Leaver is the reference, and the PINN is trained against an FD ground truth.
- The contribution is methodological: *controlled benchmark* of a class of methods on a problem with known answers.

Where it still pays:

- **Inverse problem:** jointly fit PDE + ringdown data, recover M, ω, τ in one shot.
- **Differentiable surrogate:** once trained, Φ_θ gives free derivatives w.r.t. M for downstream inference.

Open directions:

- Multi-mode inverse template (variant E started, τ to 1.30%).
- Noisy / short-window inverse: PINN vs. Prony/ESPRIT shoot-out.
- Kerr ($a \neq 0$) — two coupled equations, harder ground truth.

Thank you

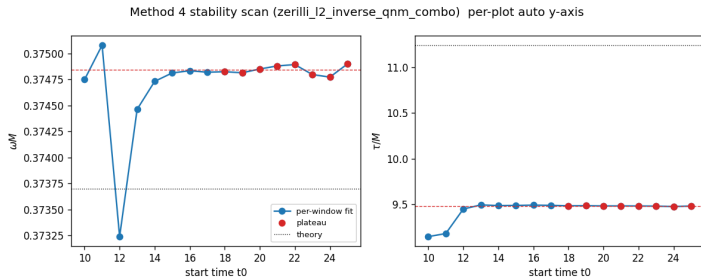
Questions?

Appendix — Inverse PINN variants (M , ω , τ recovery)

Variant	M err	ω err (M4)	τ err (M2 wide)	Change vs. baseline
Baseline	0.41%	1.46%	5.27%	—
A: $t_{\text{ring}}^{\text{min}} = 18M$	0.45%	0.89%	0.43%	later ringdown window
B: $\lambda_{\text{ring}} = 100$	1.25%	nan	23.27%	ringdown weight up
C: $N_{\text{ring}} = 1000$	0.34%	50.3%	5.14%	more template points
D: A+B+C combined	0.34%	0.31%	1.59%	headline result
E: two-mode template	1.44%	2.63%	1.30%	joint fundamental + overtone

- Single-axis tweaks (A, B, C alone) trade off: each fixes one bias and worsens another.
- Combined recipe (D) is the only configuration where all three parameters are simultaneously below 1.6%.

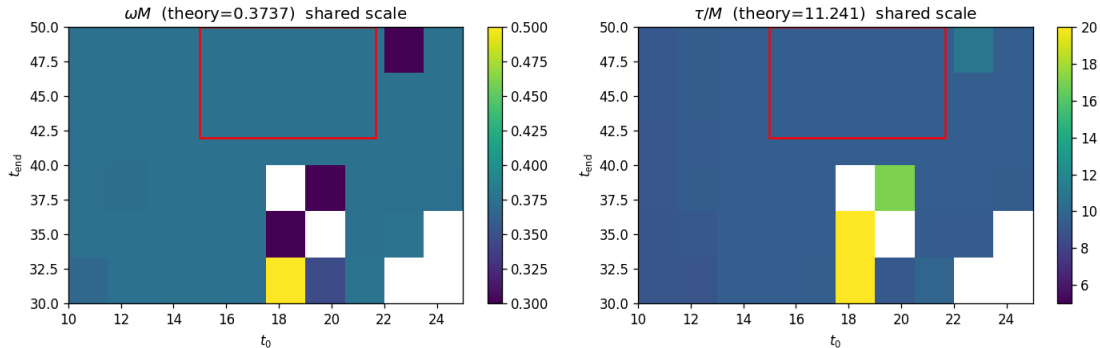
Appendix — M4 stability scan, inverse PINN



Variant D inverse PINN: M4 plateau is sharp and narrow ($\sigma_\omega = 4.4 \times 10^{-5}$).

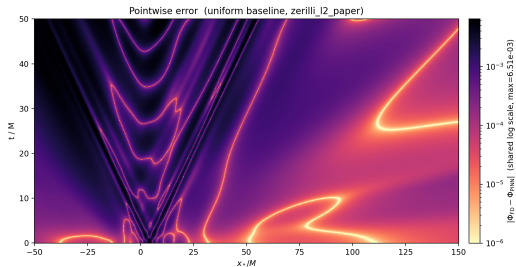
Appendix — M5 2-D heatmap, inverse PINN

Method 5 2-D stability scan (zerilli_I2_inverse_qnm_combo)

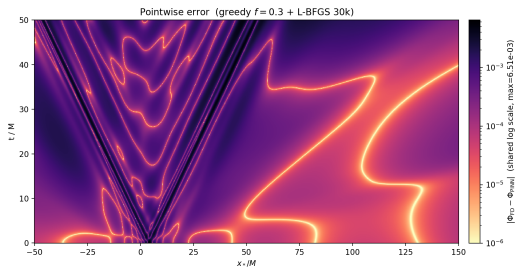


Variant D inverse PINN: M5 confirms a wide stable rectangle in the (t_0, t_{end}) plane.

Appendix — Greedy vs. baseline error heatmaps



Uniform baseline (Patel et al. 2024 reproduction).



Greedy $f=0.3$ + L-BFGS 30k (best).

Appendix — M3 (ESPRIT) algorithm

Given $y[n] = \sum_{k=1}^K A_k z_k^n$ with $z_k = e^{(i\omega_k - 1/\tau_k)\Delta t}$:

- 1 Build Hankel matrix $H \in \mathbb{C}^{(N-L) \times L}$ from y .
- 2 Truncated SVD: $H = U\Sigma V^\dagger$, keep top K columns $\Rightarrow$ signal subspace U_s .
- 3 Form $U_{s,1}$ (drop last row), $U_{s,2}$ (drop first row).
- 4 Solve $U_{s,1}\Psi = U_{s,2}$ in least squares.
- 5 Eigenvalues of Ψ are $\{z_k\} \Rightarrow \omega_k = \text{Im}(\log z_k)/\Delta t$, $\tau_k^{-1} = -\text{Re}(\log z_k)/\Delta t$.
- 6 Solve a linear least-squares for $\{A_k\}$; report largest-amplitude mode.

Conditioning check: we flag the result as ill-conditioned if the dominant amplitude falls below a numerical threshold (here $|A| < 10^{-20}$); this catches cases where ESPRIT has fitted noise.